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Advanced Computer Architecture

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Name
Last Name

Pick 2 out of th 3 problems, DO NOT DO ALL THE 3!

Questions (from Forms) (20%)	
Problem 1 (40%)	
Problem 2 (40%)	
Problem 3 (40%)	
Total (100%)	

Problem 1

Assume that the following code has been executed on a CPU with TOMASULO.

	Instruction	ISSUE	START EXE	WB
1	SUBD F2, F4, F6	1	2	5
2	MULTD F4, F8, F0	2	3	7
3	ADDD F6, F2, F4	3	8	10
4	ADDD F2, F10, F12	4	5	7
5	MULTD F14, F2, F4	5	9	13

- A. List all the possible conflicts in the code.
- B. Is there a “configuration” that can respect the shown execution?
How many units? Which kind? What latency?
- C. If the previous table was not correct, please, write the right one and specify the number, kind and latency for each unit.

Answer 1.A

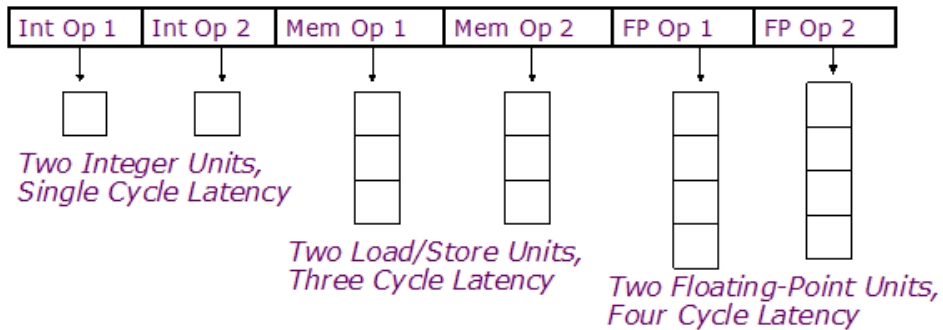
Answer 1.B

Answer 1.C

	Instruction	ISSUE	START EXE	WB
1	SUBD F2, F4, F6			
2	MULTD F4, F8, F0			
3	ADDD F6, F2, F4			
4	ADDD F2, F10, F12			
5	MULTD F14, F2, F4			

Problem 2

Considering the following VLIW “architecture”:



Considering the following portion of assembly, and schedule the following code for the VLIW, consider a single iteration.

Branch completed with 1 cycle delay slot (branch solved in ID stage)

```

LOOP: LD $F1, 0($R1)
      FADD $F2, $F2, $F3
      ADDI $R3, $R3, 8
      LD $F4, 0($R2)
      FADD $F5, $F4, $F2
      FMULT $F6, $F1, $F4
      ADDI $R5, $R5, 1
      LD $R6, 0($R4)
      SD $F6, 0($R5)
      SD $F5, 0($R6)
      BNE $R5, $R7, LOOP
  
```

Answer

	ALU1	ALU2	MU1	MU2	FPU1	FPU2
C1						
C2						
C3						
C4						
C5						
C6						
C7						
C8						
C9						
C10						
C11						
C12						
C13						
C14						
C15						
C16						
C17						
C18						
C19						
C20						

Problem 3

Design and describe (all the decisions have to be motivated) a 1-BHT and a 2-BHT able to execute the following assembly code.
(R0 is set to 100, R1 is set to 0)

```
LOOP:    LD      F3  0    R0
         ADDD    F1  F3  F3
         MULTD   F2  F3  F1
         ADDI    R1  R1  1000
LOOP2:   LD      F3  0    R1
         MULTD   F2  F2  F3
         SUBI    R1  R1  10
         BNEZ    R1  LOOP2
         SUBI    R0  R0  10
         BNE     R0  R1  LOOP
```

The obtained result, in terms of miss predictions, is inline with theoretical characteristics of the two predictors? Please effectively support your answer.

Answer

